

Sets and Functions

Applied Algebra
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Why do we need sets?

We have already used sets many times, even if not very formally.

Examples:

$$\mathbb{Z}, \quad \{\text{prime numbers}\}, \quad \{x \in \mathbb{Z} : x \equiv 1 \pmod{4}\}.$$

We will make the language precise.

Important point. Sets are basic objects in mathematics. Later, functions, relations, groups, and many other structures will all be built from them.

Examples of sets

A *set* is a collection of objects, called its *elements* or *members*.

Examples:

$$A = \{1, 2, 3, 4, 5\},$$

$$B = \{x \in \mathbb{Z} : x^2 < 10\},$$

$$C = \{\text{students in this class}\}.$$

Some remarks:

- The elements of a set can be numbers, people, matrices, functions, and so on.
- A set is determined only by its elements.
- The order of the elements does not matter.

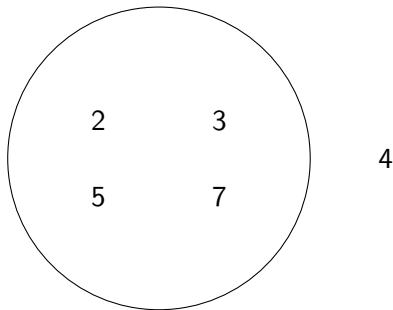
For example,

$$\{2, 3, 5\} = \{5, 2, 3\}.$$

Membership notation

If x is an element of a set X , we write $x \in X$. Otherwise, we write $x \notin X$.

Example. $2 \in \{2, 3, 5, 7\}$ while $4 \notin \{2, 3, 5, 7\}$.



How can we describe a set?

There are two common ways.

1. By listing the elements

$$X = \{2, 3, 5, 7, 11, 13\}.$$

2. By a property (set-builder notation)

$$X = \{p \in \mathbb{P} : p \text{ is prime and } p < 15\}.$$

The symbol “:” means *such that*.

So the second description reads:

X is the set of positive integers p such that p is prime and p < 15.

These two descriptions define the same set.

Listing notation vs. set-builder notation

Listing notation is convenient when the set is small and explicit.

$$\{1, 2, 3, 4, 5\}$$

Set-builder notation is better when the set is large or defined by a rule.

$$\{x \in \mathbb{Z} : -2 \leq x \leq 4\}$$

Examples.

$$\{2, 4, 6, 8, 10\} = \{2n : n = 1, 2, 3, 4, 5\},$$

$$\{0, \pm 1, \pm 2, \pm 3, \dots\} = \mathbb{Z},$$

$$\{1, 4, 9, 16, 25, \dots\} = \{n^2 : n \in \mathbb{N}, n \geq 1\}.$$

In practice, we choose whichever description is clearer.

Examples: convert between the two forms

Rewrite the following sets in a different form.

Example 1. Write by listing:

$$A = \{x \in \mathbb{Z} : 1 \leq x \leq 5\}.$$

Answer: $A = \{1, 2, 3, 4, 5\}$.

Example 2. Write by a property:

$$B = \{2, 4, 6, 8, 10\}.$$

One possible answer is $B = \{2n : n = 1, 2, 3, 4, 5\}$.

Another is $B = \{x \in \mathbb{N} : x \text{ is even and } 2 \leq x \leq 10\}$.

Subset

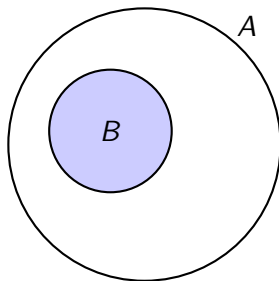
Let A and B be sets. We say that B is a **subset** of A , and write

$$B \subseteq A,$$

if every element of B is also an element of A .

Examples.

- $\{2, 3\} \subseteq \{1, 2, 3, 4\}$, $\{\text{even integers}\} \subseteq \mathbb{Z}$.
- $\{1, 5\} \not\subseteq \{1, 2, 3, 4\}$.
- $A \subseteq A$.



Equality of sets

Two sets A and B are equal if they contain exactly the same elements.

This means:

- every element of A lies in B , and
- every element of B lies in A .

Hence

$$A = B \quad \Longleftrightarrow \quad A \subseteq B \text{ and } B \subseteq A.$$

Examples.

- $\{1, 2, 3\} = \{3, 2, 1\}$, because order does not matter.
- $\{x \in \mathbb{Z} : x^2 = 1\} = \{-1, 1\}$.

Union

Let A and B be sets. The **union** of A and B is

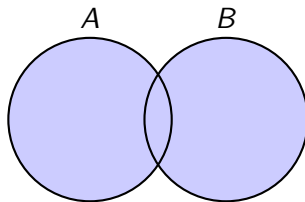
$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

Example.

$$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\}.$$

Then

$$A \cup B = \{1, 2, 3, 4, 5\}.$$



Remark. An element is in $A \cup B$ if it is in *at least one* of the sets.

Intersection

The **intersection** of A and B is

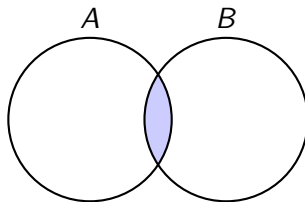
$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

Example.

$$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\}.$$

Then

$$A \cap B = \{3\}.$$



Remark. An element is in $A \cap B$ only if it belongs to *both* sets.

Difference

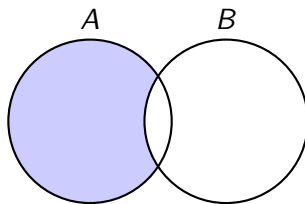
The **difference** of A and B is

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

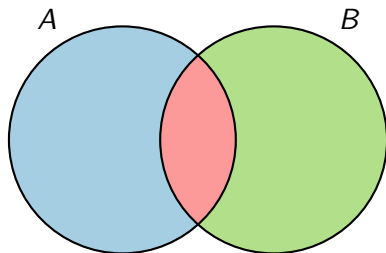
Example. Consider $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$.
Then

$$A \setminus B = \{1, 2\}, \quad B \setminus A = \{5\}.$$

Remark. Order matters: in general $A \setminus B \neq B \setminus A$.



Venn diagram



Legend.

- Blue: $A \setminus B$ (only in A)
- Green: $B \setminus A$ (only in B)
- Red: $A \cap B$ (in both)

A key identity

We have the distributive law:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Remark. This is similar to distributivity in algebra.

Why do we need functions?

Sets describe collections of objects.

Very often, however, mathematics is not only about objects, but also about *rules* connecting one collection to another.

Examples:

- each integer x is sent to its square x^2 ;
- each real number x is sent to $x + 1$;
- each student is sent to their student ID number;
- each integer is sent to its remainder modulo 5.

These are all examples of **functions**.

Definition of a function

Let A and B be sets. A function f from A to B is a rule which assigns to each element $a \in A$ a unique element $f(a) \in B$.

We write

$$f : A \rightarrow B.$$

Here:

- A is the **domain**;
- B is the **codomain**;
- $f(a)$ is the value of the function at a .

Key requirement: for every $a \in A$, there is exactly one element $f(a) \in B$.

Remark. A function is not just a formula. It also includes the domain and codomain.

Examples of functions

Example 1.

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(x) = x + 1.$$

Each integer is sent to the next integer.

Example 2.

$$g : \mathbb{R} \rightarrow \mathbb{R}, \quad g(x) = x^2.$$

Each real number is sent to its square.

Example 3.

$$h : \mathbb{Z} \rightarrow \{0, 1, 2, 3, 4\},$$

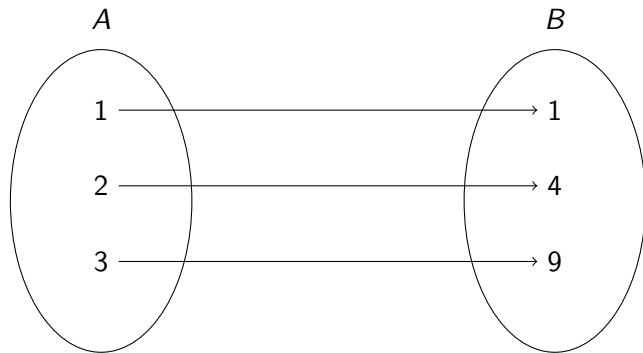
where $h(n)$ is the remainder when n is divided by 5.

For instance,

$$h(7) = 2, \quad h(11) = 1, \quad h(-3) = 2.$$

These are all functions because each allowed input has exactly one output.

A mapping diagram



This picture represents the function

$$f : \{1, 2, 3\} \rightarrow \{1, 4, 9\}, \quad f(x) = x^2.$$

Each element of the domain has exactly one arrow leaving it.

Domain and codomain

Suppose

$$f : A \rightarrow B.$$

Then:

- the **domain** is the set A of allowed inputs;
- the **codomain** is the set B , where outputs are supposed to lie.

Example.

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

Then:

- domain = $\mathbb{R}$;
- codomain = $\mathbb{R}$.

If instead we write

$$f : \mathbb{R} \rightarrow [0, \infty), \quad f(x) = x^2,$$

then the formula is the same, but the codomain is different.

Image of a function

Let

$$f : A \rightarrow B.$$

The **image** of f is the set of actual outputs:

$$f(A) = \{f(a) : a \in A\}.$$

So the image is always a subset of the codomain:

$$f(A) \subseteq B.$$

Example. Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

Then the codomain is $\mathbb{R}$, but the image is

$$f(\mathbb{R}) = [0, \infty).$$

A common mistake: codomain is not the same as image

Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

Students often say:

“the codomain is $[0, \infty)$.”

That is **wrong** in this example.

Here:

- the **codomain** is $\mathbb{R}$, because that is what appears after the arrow;
- the **image** is $[0, \infty)$, because those are the actual outputs.

So:

$$\text{image} \subseteq \text{codomain},$$

but they need not be equal.

Injective functions

Let $f : A \rightarrow B$. We say that f is **injective** if different inputs always give different outputs.

In symbols:

$$f(a_1) = f(a_2) \implies a_1 = a_2.$$

Equivalently,

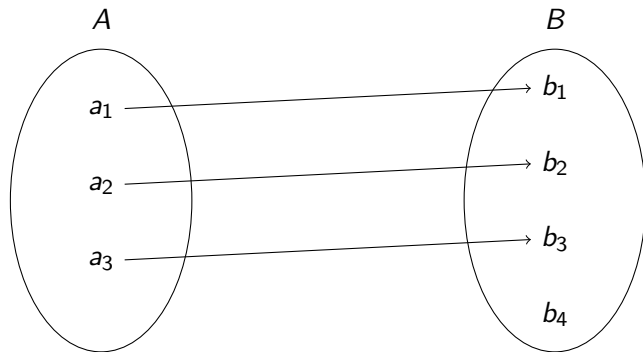
$$a_1 \neq a_2 \implies f(a_1) \neq f(a_2).$$

Informal meaning. No two distinct elements of the domain are sent to the same element of the codomain.

Short slogan.

Injective = no collisions.

Injective: mapping diagram



This function is injective: different elements of A go to different elements of B .

It is allowed that some elements of B are not hit.

Examples of injective functions

Example 1.

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(x) = x + 1.$$

This is injective, because

$$x + 1 = y + 1 \implies x = y.$$

Example 2.

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = 2x - 3.$$

This is injective, because

$$2x - 3 = 2y - 3 \implies x = y.$$

Example 3.

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(n) = 2n.$$

This is injective: distinct integers have distinct doubles.

A non-example of injectivity

Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

This is **not** injective.

Indeed,

$$f(1) = 1, \quad f(-1) = 1,$$

but

$$1 \neq -1.$$

So two different inputs give the same output.

Conclusion.

$$f(x) = x^2 \text{ on } \mathbb{R} \text{ is not injective.}$$

But: if we restrict the domain to $[0, \infty)$, then the same rule

$$f : [0, \infty) \rightarrow \mathbb{R}, \quad f(x) = x^2$$

is injective.

Surjective functions

Let $f : A \rightarrow B$. We say that f is **surjective** if every element of the codomain is hit by the function.

In symbols:

$$\forall b \in B, \exists a \in A \text{ such that } f(a) = b.$$

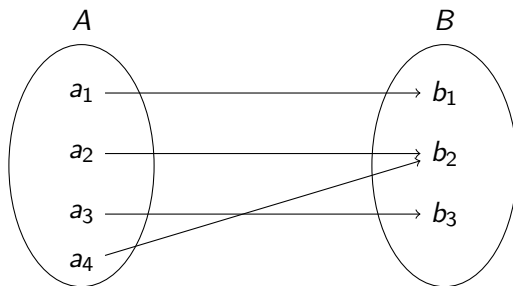
Equivalently, the image is the whole codomain:

$$f(A) = B.$$

Informal meaning. Every element of B occurs as an output.

Examples of surjective functions

Example 1.



This function is surjective: every element of B has at least one arrow coming to it.

It is allowed that several elements of A go to the same element of B .

Example 2.

$$f : \mathbb{R} \rightarrow [0, \infty), \quad f(x) = x^2.$$

This is surjective, because every $y \geq 0$ can be written as

$$y = x^2$$

for some real number x , namely $x = \sqrt{y}$ or $x = -\sqrt{y}$.

Example 3.

$$f : \mathbb{Z} \rightarrow \{0, 1, 2\},$$

where $f(n)$ is the remainder of n upon division by 3.

This is surjective, since each of 0, 1, 2 occurs as an output.

For example:

$$f(0) = 0, \quad f(1) = 1, \quad f(2) = 2.$$

A non-example of surjectivity

Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

This is **not** surjective.

Why? Because no real number x satisfies

$$x^2 = -1.$$

So -1 is in the codomain, but it is not in the image.

Therefore,

$$f(\mathbb{R}) = [0, \infty) \neq \mathbb{R}.$$

Bijjective functions

A function is called **bijjective** if it is both injective and surjective.

So a bijection has two properties:

- different inputs give different outputs;
- every element of the codomain is hit.

Informal meaning. A bijection gives a perfect matching between the domain and the codomain.

Bijjective functions are especially important, because they tell us that two sets have exactly the same size in the strongest possible sense.

Examples of bijections

Example 1.

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(x) = x + 1.$$

This is bijective:

- injective, because $x + 1 = y + 1$ implies $x = y$;
- surjective, because for every $m \in \mathbb{Z}$, we have $m = f(m - 1)$.

Example 2.

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = 2x + 1.$$

This is bijective:

- injective;
- surjective, since for every $y \in \mathbb{R}$,

$$y = f\left(\frac{y - 1}{2}\right).$$

Quick check: classify these functions

Determine whether each function is injective, surjective, or bijective.

(a)

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2$$

(b)

$$g : [0, \infty) \rightarrow [0, \infty), \quad g(x) = x^2$$

(c)

$$h : \mathbb{Z} \rightarrow \mathbb{Z}, \quad h(n) = n + 5$$

(d)

$$k : \mathbb{Z} \rightarrow \mathbb{Z}, \quad k(n) = 2n$$

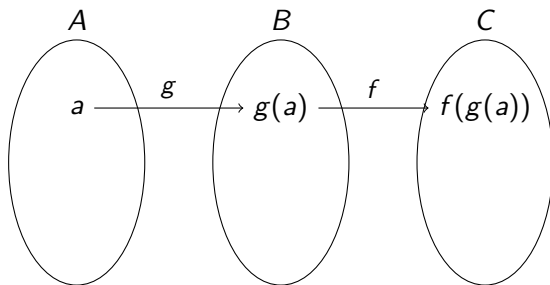
Composition of functions

$$g : A \rightarrow B, \quad f : B \rightarrow C$$

$$f \circ g : A \rightarrow C$$

$$(f \circ g)(x) = f(g(x))$$

First g , then f



$$a \mapsto g(a) \mapsto f(g(a))$$

Composition: example

Let

$$\begin{aligned}g : \mathbb{R} &\rightarrow \mathbb{R}, & g(x) &= x + 1, \\f : \mathbb{R} &\rightarrow [0, \infty), & f(x) &= x^2.\end{aligned}$$

Then

$$(f \circ g)(x) = f(g(x)) = (x + 1)^2.$$

Remark. Let $f, g : A \mapsto A$. In general,

$$f \circ g \neq g \circ f.$$

Identity function

For a set A , the **identity function** is

$$\text{id}_A : A \rightarrow A, \quad \text{id}_A(x) = x.$$

Property. For any function $f : A \rightarrow B$,

$$f \circ \text{id}_A = f, \quad \text{id}_B \circ f = f.$$

Left inverse and injectivity

Let $f : A \rightarrow B$. A function $g : B \rightarrow A$ is a **left inverse** of f if

$$g \circ f = \text{id}_A.$$

In the above formula, g is in the left hand side of f .

Interpretation.

$$g(f(x)) = x \quad \text{for all } x \in A$$

Key fact.

$$f \text{ has a left inverse} \iff f \text{ is injective.}$$

Idea. If $f(x_1) = f(x_2)$, then applying g gives $x_1 = x_2$.

Right inverse and surjectivity

Let $f : A \rightarrow B$. A function $h : B \rightarrow A$ is a **right inverse** of f if

$$f \circ h = \text{id}_B.$$

In the above formula, h is in the right hand side of f .

Interpretation.

$$f(h(y)) = y \quad \text{for all } y \in B$$

Key fact.

$$f \text{ has a right inverse} \iff f \text{ is surjective.}$$

Idea. For each $y \in B$, pick $h(y)$ with $f(h(y)) = y$.

Inverse functions

Let $f : A \rightarrow B$. Its **inverse** is a function $f^{-1} : B \rightarrow A$ such that

$$f^{-1} \circ f = \text{id}_A, \quad f \circ f^{-1} = \text{id}_B.$$

Key fact.

$$f \text{ invertible} \iff f \text{ bijective.}$$

Remark.

- left inverse + right inverse $\Rightarrow$ they coincide
- unique inverse: f^{-1}

Inverse: example

Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = 2x + 1.$$

To find the inverse, solve

$$y = 2x + 1.$$

Then

$$x = \frac{y - 1}{2}.$$

So

$$f^{-1}(y) = \frac{y - 1}{2}.$$

Check.

$$f^{-1}(f(x)) = x.$$

A non-example: no inverse

Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

This function is not injective.

Indeed,

$$f(1) = f(-1).$$

So if we try to define an inverse,

$$f^{-1}(1) = ?$$

There are two possible answers: 1 and -1 .

Conclusion. No inverse exists.

Functions and sets

Functions describe relationships between sets.

Viewpoint.

- sets = objects;
- functions = structure between objects.

This viewpoint will be very important later (especially in group theory).

Key idea. Understanding functions means understanding how mathematical objects are related.

Bijections and size

If there exists a bijection

$$f : A \rightarrow B,$$

then A and B have the same size.

This is how we compare sizes of sets in mathematics.

Example.

$$f : \mathbb{Z} \rightarrow 2\mathbb{Z}, \quad f(n) = 2n.$$

This is a bijection.

So the set of integers and the set of even integers have the same size.

This may seem surprising, but it is very important.

Final practice

(1) Compute:

$$(f \circ g)(x), \quad (g \circ f)(x),$$

where

$$f(x) = x^2, \quad g(x) = x + 2.$$

(2) Determine whether the function is invertible:

$$f(x) = x^3, \quad f : \mathbb{R} \rightarrow \mathbb{R}.$$

(3) Find the inverse (if it exists):

$$f(x) = 3x - 5.$$